A Survey on Cascaded Multilevel Inverters

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***Abstract*—Cascaded multilevel inverters synthesize a medium-voltage output based on a series connection of power cells which use standard low-voltage component configurations. This characteristic allows one to achieve high-quality output voltages and input currents and also outstanding availability due to their intrinsic component redundancy. Due to these features, the cascaded multilevel inverter has been recognized as an important alternative in the medium-voltage inverter market. This paper presents a survey of different topologies, control strategies and modulation techniques used by these inverters. Regenerative and advanced topologies are also discussed. Applications where the mentioned features play a key role are shown. Finally, future developments are addressed.**

***Index Terms*—Cascaded inverters, modulation and control, multilevel converters.**

I. Introduction

MULTILEVEL voltage-source inverters provide a cost-effective solution in the medium-voltage energy management market [1]. These converters have been widely applied to chemical, oil, and liquefied natural gas (LNG) plants, water plants, marine propulsion, power generation, energy transmission, and power-quality devices [2].

Nowadays, there exist three commercial topologies of multilevel voltage-source inverters: neutral point clamped (NPC) [3], cascaded H-bridge (CHB) [4], and flying capacitors (FCs) [5]. Among these inverter topologies, cascaded multilevel inverter reaches the higher output voltage and power levels (13.8 kV, 30 MVA) and the higher reliability due to its modular topology.

Cascaded multilevel inverters are based on a series connection of several single-phase inverters. This structure is capable of reaching medium output voltage levels using only standard low-voltage mature technology components. Typically, it is necessary to connect three to ten inverters in series to reach the required output voltage.

These converters also feature a high modularity degree because each inverter can be seen as a module with similar circuit topology, control structure, and modulation [6]. Therefore, in the case of a fault in one of these modules, it is possible to replace it quickly and easily. Moreover, with an appropriated control strategy, it is possible to bypass the faulty module without stopping the load, bringing an almost continuous overall availability [7].

This paper presents a bibliographical review of cascaded multicell inverters, its working principle, circuit topologies, control techniques, and industrial applications. This paper is organized as follows. Section II shows the working principle and basic and advanced topologies. Regenerative topologies and its control are addressed in Section III. In Section IV, several newly introduced topologies for cascaded inverters are shown. A complete review of control and modulation used in these inverters is presented in Section V. A group of applications is reviewed in Section VI. Finally, future trends and conclusions are presented in Sections VII and VIII, respectively.

II. Power Circuit Topologies

A basic structure of a cascaded multilevel inverter is shown in Fig. 1(a). Each inverter uses a dc-link voltage to generate a modulated voltage at the output terminals. The total output voltage is obtained by the sum of each individual output voltage as shown in Fig. 1(b). Each inverter is able to produce three output voltage levels, namely, $+v_{\rm dc}$, $-v_{\rm dc}$, and 0. The maximum number of voltage levels of the phase voltage $L_{\rm ph}$ is given by

$$L_{\rm ph} = 2N_{\rm inv} + 1 \tag{1}$$

where $N_{\rm inv}$ is the number of inverters.

Each inverter requires an isolated dc voltage which is usually obtained by an arrangement of three-phase or single-phase rectifiers [8], as shown in Fig. 2, and a multipulse transformer which provides the electrical isolation. In some applications, these dc voltages can be obtained directly by isolated dc sources, for example, photovoltaic panels [9] or dc/dc isolated converters [10]. In another applications, like STATic COMpensator (STATCOM), which does not require the injection of active power, the dc voltages can be floating, and the control strategy keeps the dc-link voltage adjusted to the reference [11].

A further optimization in terms of input current harmonics can be done when a multipulse transformer is used to provide the isolated dc sources. By using a different phase angle for each group of secondaries, i.e., secondaries that fed the inverters of each output phase, it is possible to eliminate characteristic harmonics produced by the diode-based rectifiers [4]. These phase angles can be calculated using

$$\Delta\varphi = \frac{\pi}{3N_{\rm inv}} \tag{2}$$

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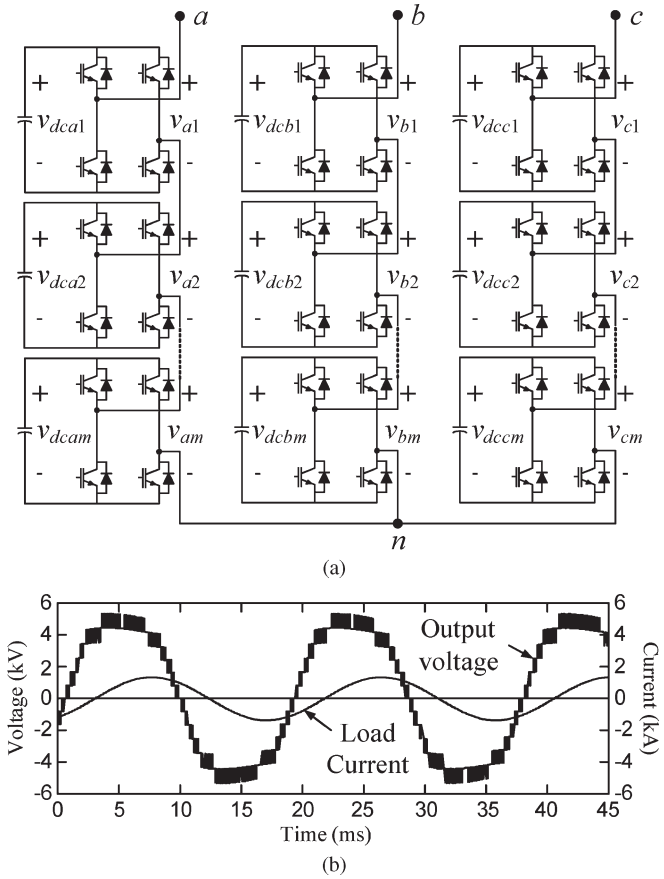


Fig. 1. Multilevel cascaded inverter. (a) Basic circuit topology. (b) Typical multilevel output voltage and load current.

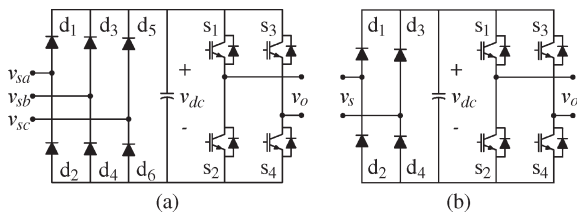


Fig. 2. Diode-based rectifier power cell topologies. (a) Three-phase. (b) Single-phase.

where $\Delta\varphi$ is the relative angle between group of secondaries.

When single phase rectifiers are used the secondaries must be arranged in a more complex structure [8].

The use of a cascaded five-level NPC inverter instead of a three-level inverter has received an increased attention [12]–[15], due to the possibility to increase the total number of output voltage levels by keeping a fixed number of inverters. Moreover, the NPC topology has become an established technology in power electronics inverters. The total number of levels using cascaded five-level NPC inverters is given by

$$L_{\mathrm{ph}}^{\mathrm{NPC}} = 4N_{\mathrm{NPC}} + 1 \tag{3}$$

where N_{NPC} is the number of NPC inverters connected in series.

By using the same principle, there is a possibility to connect in series two or more FC inverters in order to increase the number of output voltage levels. In [16], two FC inverters are connected in series to obtain a 13-level output voltage.

Usually, cascaded multilevel inverters use the same dc-link voltage value for every cell. However, using different dc-link voltages, it is possible to increase the maximum number of output voltage levels. The topologies that have different dc-link voltages are called in the literature as asymmetric cascaded inverter. The relationship among the dc-link voltages to provide a regularly stepped output voltage waveform could be binary (power of two) or trinary (power of three). The maximum numbers of output voltage levels are given by

$$L_{\mathrm{ph}}^{\mathrm{bin}} = 2^{(N_{\mathrm{inv}}+1)} - 1 \quad L_{\mathrm{ph}}^{\mathrm{tri}} = 3^{N_{\mathrm{inv}}}. \tag{4}$$

The asymmetric inverter configuration in Fig. 3 shows a cascaded-inverter-driven induction machine using trinary asymmetry and three inverters per phase [17], [18]. It is possible to note that, although the dc voltages have a proportion of 9/3/1, the nominal power driven by each inverter has a different proportion, where the high-voltage cell manages the major part (89%) of the output power [19]. Due to this power distribution, these kinds of inverters can be designed with different switch technologies like integrated gate commutated thyristor for high-power inverters, high-voltage insulated-gate bipolar transistor for medium-power inverters, and low-voltage insulated-gate bipolar transistor for low-power inverters. Cascaded multilevel inverters that use different switch technologies are also called in the literature as hybrid cascaded inverters [20]. One of the main drawbacks of this topology is the loss of modularity produced due to the different semiconductor technologies used. Problems with switching frequency and modulation index restrictions produced in this topology have also been addressed [17].

As mentioned before, when the inverter does not need to deliver active power, like in the STATCOM application, it is possible to use only dc capacitors instead of isolated dc sources. This configuration is also possible when one of the cascaded inverters is producing the total output active power and the other inverters are used as a support to improve the output voltage waveform. One of the best suited topologies to use floating cells is the asymmetric converter [21], [22], where the high-voltage inverter can positively deliver the entire output power and the low-voltage inverters are used with floating dc links [23]. Alternative configurations use a combination of two-level [24] or three-level NPC [25] three-phase inverter with cascaded single-phase inverters as shown in Fig. 4. The topology with an NPC converter in series with a floating single-phase inverter with floating dc sources shown in [26] offers an optimum tradeoff between output quality, reliability and efficiency.

Recently, the use of a multilevel dc-link voltage has been studied in order to increase the number of total output voltage levels using only a few single-phase inverters. The topology is based on a variable dc-link voltage which could have zero to several voltage levels; then, a single-phase full-bridge (SPFB) inverter could apply this voltage or its negative. These topologies could deliver a high output current with low switching frequency in the SPBF. The variable dc-link voltage can be

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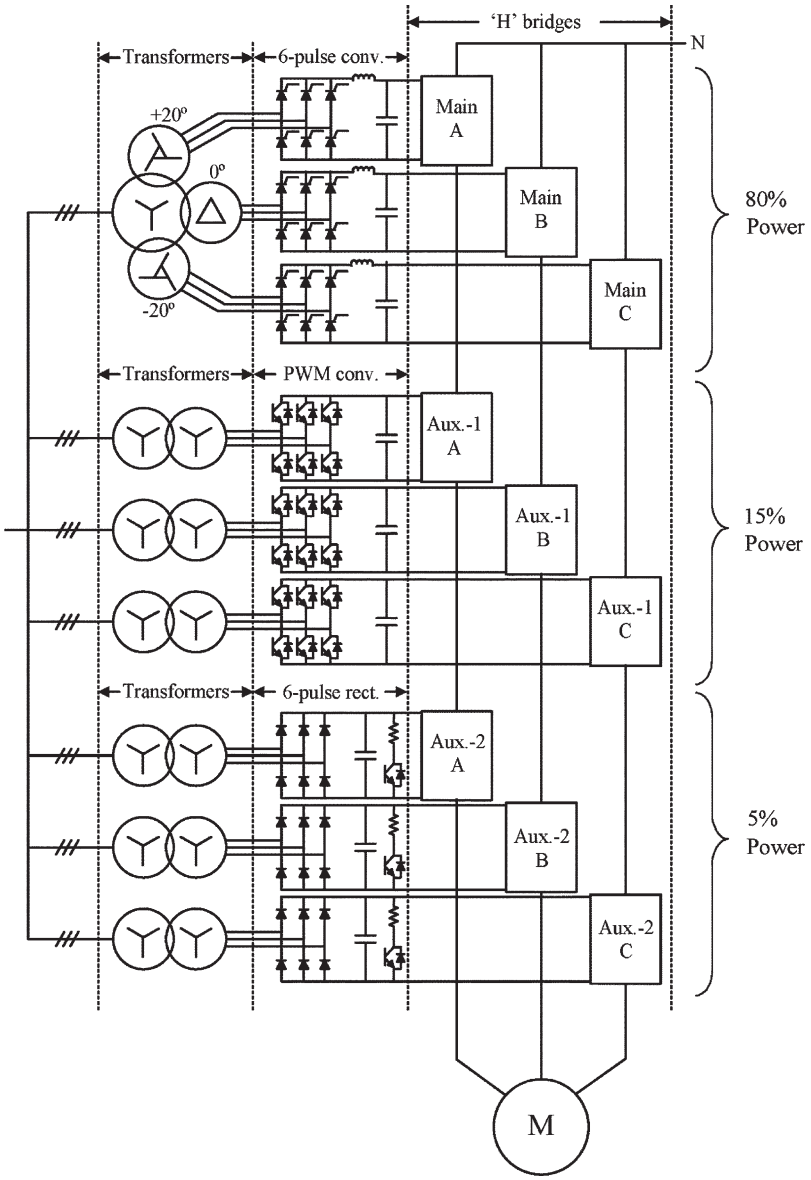


Fig. 3. Asymmetric/hybrid cascaded circuit topology.

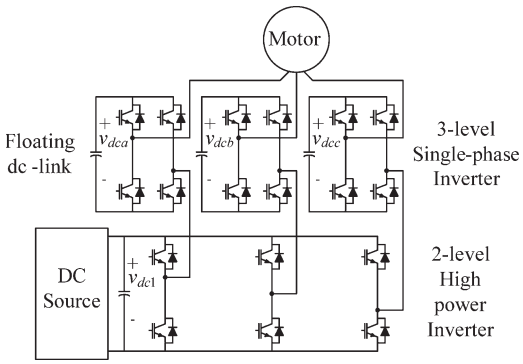


Fig. 4. Cascaded circuit topologies with floating dc link.

produced by cascaded NPC or FC configurations or even buck dc–dc converters [27], [28]. Fig. 5 shows two of these configurations.

III. Regenerative Topologies

The standard topology with a diode-based rectifier is extensively used with pumps and fans applications where the regenerative issues are minimal and can be managed by resistive damping. However, several applications like downhill conveyors, elevators, and energy plants require operations with a bidirectional power flow. In those applications, classical cascaded multilevel inverters cannot provide the required regenerative operation. In order to increase the applicability of cascaded multilevel inverters, regenerative topologies have been proposed recently [29]–[32].

To achieve the regenerative operation, the diode-based rectifier must be changed by an active front-end rectifier, as shown in Fig. 6(a). Although these rectifiers require an additional controller, they have several attractive features, namely, regenerative operation, independent control of the active and reactive powers, precise control of the dc voltage, and smaller harmonic content than that of the diode-based rectifiers in the input

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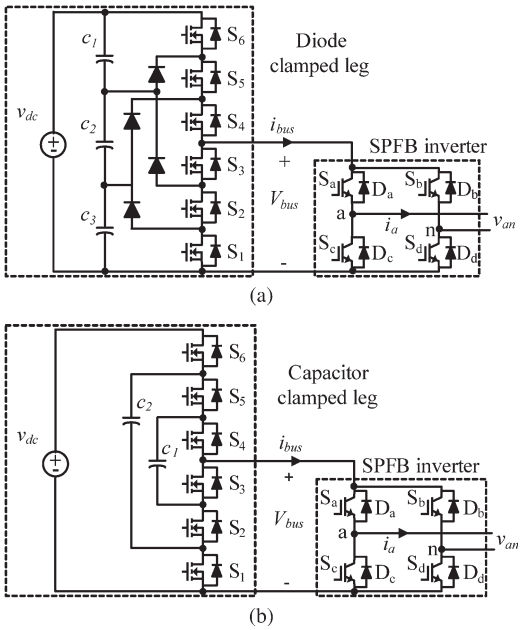


Fig. 5. Cascaded inverter with variable multilevel dc-link voltage. (a) Diode clamped. (b) Capacitor clampled.

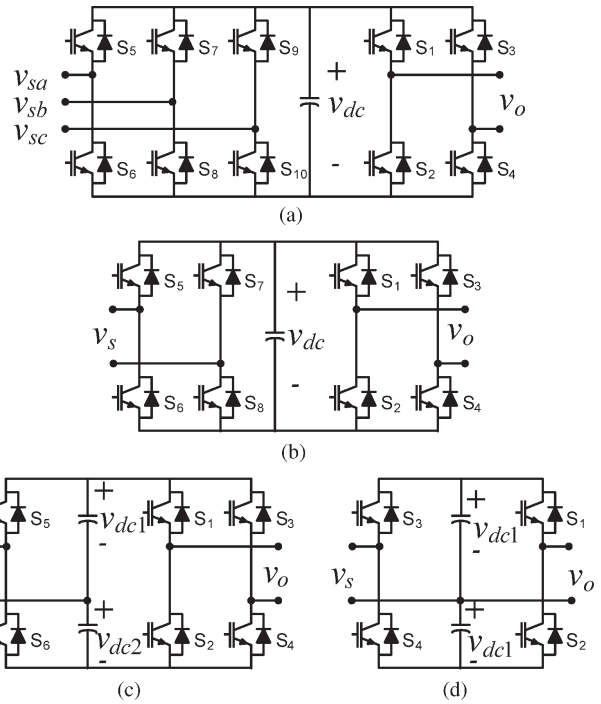


Fig. 6. Regenerative cascaded inverter topologies. (a) Three-phase. (b) Single-phase. (c) Semireduced. (d) Reduced.

current. Moreover, the active front-end rectifiers do not produce low-order harmonics in the input current, and therefore, they do not require the secondary phase angle, simplifying the transformer design.

The regenerative drive could use a three-phase [29] or single-phase [30] active rectifier as it is shown in Fig. 6(a) and (b), respectively. It is possible to merge the rectifier and inverter stage to produce the topologies shown in Fig. 6(c) and (d) which have a reduced number of switches and less related electronics [31]. Specifically, the topology shown in Fig. 6(d) uses only

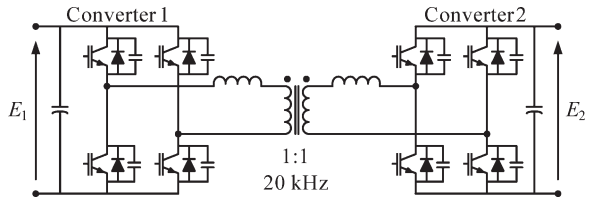


Fig. 7. High-frequency transformer-based power module to be used in cascaded inverters.

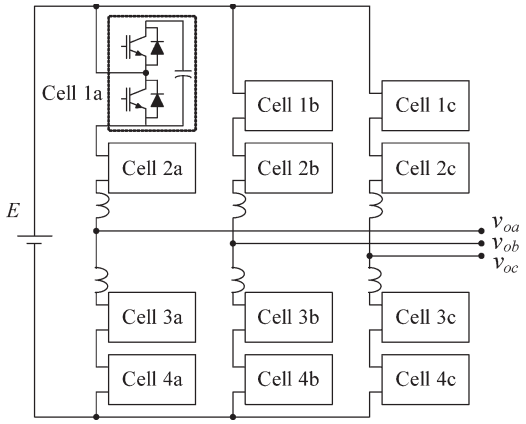


Fig. 8. MMC based on cascaded modules.

four switches per power cell. The main drawback of the reduced topologies is the increase complexity of the control system [32].

By using a regenerative rectifier in each cell, the complete drive acts as a regenerative multilevel converter [33] managing bidirectional power flow. By using a proper controller, it is possible to decouple and control the active and reactive input powers. Compensation techniques allow the reduction of dc-link voltage ripple [34], improving the output voltage waveform. Moreover, pulsewidth modulation (PWM) using phase-shifted carriers reduces significantly the low-order input current harmonics.

IV. Advanced Circuit Topologies

There are a number of alternative topologies based on the basic principle of cascaded single-phase inverter proposed in the literature. Three of them are discussed in this section.

The first approach presented is the use of a high-frequency transformer-based module used to build cascaded multilevel converters [35] without the need of isolated dc links. The power module, shown in Fig. 7, uses a single-phase rectifier, a high-frequency transformer, and a single-phase inverter. These modules are intended to be connected in series to handle medium voltage levels (6.6 kV) in distributed generation based on renewable energy and fuel cells. The main features of this topology are the inherent galvanic isolation, bidirectional power flow, reduction of the transformer size and weight, and high efficiency.

The second topology is the modular multilevel converter (MMC) shown in [36], where several dc/dc modules with floating dc link are connected in series to obtain a single-phase or three-phase output voltage. Fig. 8 shows the complete topology of a single-phase inverter and the basic module used. If the input

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voltage is directly an ac voltage, then the basic module must be a single-phase inverter. The MMC provides high scalability, it has low filter requirements and, in addition, it does not require an input transformer. To control the input/output currents and the floating dc voltages, a scheme composed by linear PI controllers and switching logic has been proposed in [37].

The third circuit topology, recently found in the literature, is the cascaded multilevel inverter based on single-phase current-source inverters [38]. This topology is the dual of the standard voltage-source topology but uses an inductive dc link. The inductors of each one of the three output phases are magnetically coupled, minimizing the second-order harmonics that appear on the dc link. Thus, the ripple in the input currents, as well as the dc inductor footprint, is reduced.

V. Modulation and Control

This section presents a review of the most commonly used modulation techniques and control strategies in cascaded multilevel inverters.

A. Modulation Techniques

Several modulation techniques have been proposed for cascaded multilevel inverters. A high number of power electronic devices and switching redundancies bring a higher level of complexity compared with a two-level inverter counterpart. However, this complexity could be used to add additional capabilities to the modulation technique, namely, reducing the switching frequency, minimizing the common-mode voltage, or balancing the dc voltages.

Modulation techniques for cascaded multilevel inverters are usually an extension of the two-level modulations [39]. According to their switching frequency, they can be classified as follows [40]: 1) fundamental switching frequency, where each inverter has only one commutation per cycle, for example, multilevel selective harmonic elimination (SHE), space vector control, and nearest voltage level, and 2) high switching frequency, where each inverter has several commutations per cycle, for example, multilevel PWM and space vector modulation (SVM).

1) Multilevel SHE: SHE techniques can be applied to cascaded multilevel inverters using two approaches.

The first one is to consider one commutation angle per inverter; thus, the number of harmonics that can be eliminated is $N_{\text{inv}} - 1$. The switching pattern of multilevel SHE can be obtained by solving a similar set of equations to two-level SHE [41]. Numeric mathematical methods used to solve these equations are Newton, resultant theory [42], and genetic algorithms [43]. The typical waveform obtained by this technique is shown in Fig. 9(a). In these waveforms, it is possible to note that there exists a high difference among the conducting times, which produces an unbalanced power distribution. If a multipulse transformer is used, this power unbalance can lead to a distorted input current. In [43], this effect is reduced by a simple change of conducting angles. This modulation technique can be applied to symmetrical inverters when the number of output voltage levels is high or when the inverter has nonequal dc links [44].

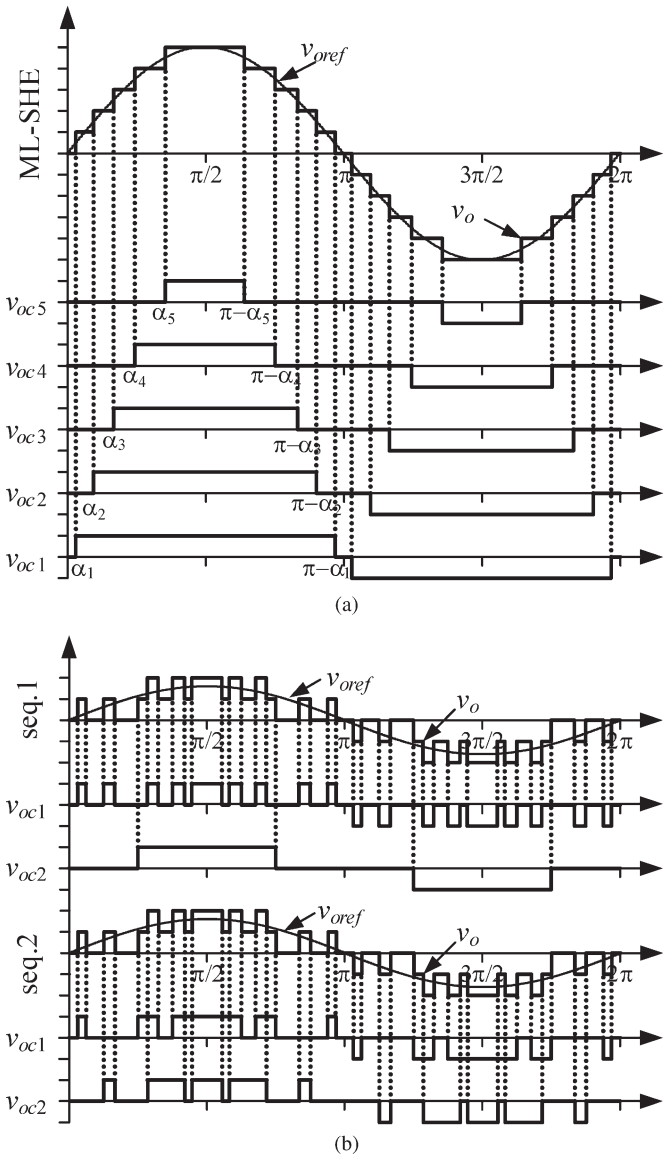


Fig. 9. Multilevel selective harmonic elimination techniques. (a) One angle per voltage level. (b) Multiple angles per voltage level.

The second approach is to combine the original SHE with the multilevel version [45] as it can be seen on the waveform of Fig. 9(b) where there are several switching angles per voltage level. In this case, the number of harmonics eliminated is independent from the number of output voltage levels, and the switching frequency is higher than the fundamental. It is possible to note that there are several different possibilities to synthesize the output voltage, allowing a further optimization in terms of switching frequency. In Fig. 9(b), the seq. 1 produces a high switching frequency in cell 2 but a fundamental switching frequency in cell 1. Alternatively, seq. 2 produces the same output voltage, but each cell has the same switching frequency.

2) Multilevel Carrier-Based PWM: Multilevel carrier-based PWM uses several triangular carrier signals, which can be modified in phase and/or vertical position in order to reduce the output voltage harmonic content. There are two common carrier modifications applied to these multilevel inverters.

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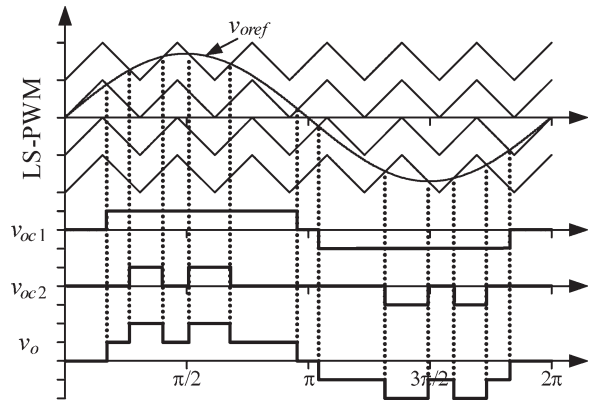


Fig. 10. Multilevel level-shifted carrier-based techniques.

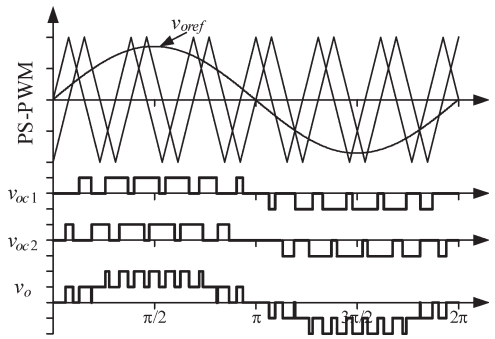


Fig. 11. Multilevel phase-shifted carrier-based techniques.

Level-shifted PWM is widely used in NPC inverters and can also be used in cascaded inverters. In [46], it is shown that this modulation technique is applied to a five-level inverter. This modulation technique produces an uneven distribution of power among cells, such as that in Fig. 10, which produces a high harmonic content in the input current. In [46], this drawback is avoided using a rotating carrier, which balances the power of each cell. In [13], the level-shifted modulation is used inside each NPC inverter and synchronized with the other cells to produce the multilevel output voltage.

Phase-shifted PWM is the most commonly used modulation technique for cascaded multilevel inverters because it offers an evenly power distribution among cells and it is very easy to implement independently of the number of inverters [47], [48]. This modulation shifts the phase of each carrier in a proper angle to reduce the harmonic content of the output voltage, as shown in Fig. 11. Moreover, it is possible to work in the overmodulation region when a common-mode term is added to the reference.

3) Multilevel SVM: Multilevel converters have a large number of vector states which can be used to modulate the reference. Moreover, each state vector has a number of redundancies, as shown in Fig. 12. Multilevel SVM must take care of this behavior to optimize the search of the modulating vectors and to apply an appropriate switching sequence [49], [50]. However, the same properties of state and switching redundancy allow the improvement of the modulation technique to fulfill additional objectives like reducing the common-mode output voltage [51], reducing the effect of overmodulation on the output currents [52], improving the voltage spectrum and minimizing the switching frequency [53], and controlling the dc-link voltage when floating cells are used [54], [55].

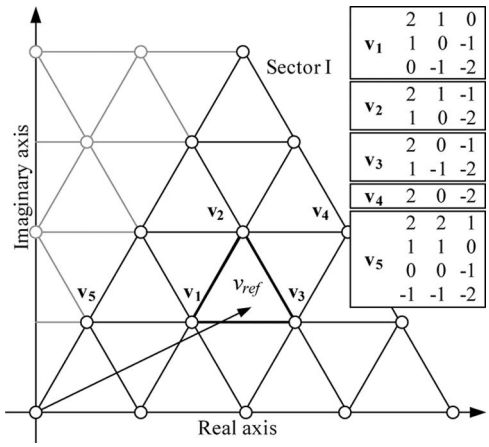


Fig. 12. Multilevel SVM.

B. Classification of Control Strategies

The control of a cascaded inverter depends mainly on its circuit topology. When a diode-based rectifier is used to generate the dc-link voltages, only the output current must be controlled. If any of the dc-link voltages is floating, a control for that voltage balance is required. In regenerative topologies, an additional controller for the input current and dc-link voltage is required. In this section, a review of different control strategies applied to cascaded multilevel inverters is shown.

In [34], a dc-link voltage compensator is proposed. The dc-link voltages are feedback to the controller modifying the reference in order to compensate its deviations. This control strategy is well suited to cascaded inverters due to the high dc-voltage ripple produced by the single-phase inverter operation. Rotating dq-transform-based controllers [56] have a simplified analysis and design because they can be a linear PI work on dc variables. However, this control scheme requires a robust synchronization method to implement the rotating transform.

To control the ac current with a high bandwidth and without requiring a synchronous transformation, it is possible to use a resonant control [30]. This controller has the restriction of constant frequency operation; therefore, it could be applied to control the input current in a regenerative topology.

Control strategies based on passivity [57] could be used to control both input and output currents. This approach offers linear and nonlinear controllers which can mathematically prove stability [58].

Recently, predictive control has been applied to cascaded inverters [59]–[61]. Predictive control considers the inverter as a system with a finite number of switching states and, using a suitable model, finds a state that optimizes a given objective function. In the previously cited references, this cost function contains terms related to the output current error and the switching frequency minimization. The main drawback of predictive control when applied to multilevel converters is the high number of possible switching states that must be evaluated.

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TABLE I
SUPPLIERS OF CHB INVERTERS AVAILABLE IN THE MARKET

Manufacturer	ABB	Siemens	TMEIC GE (1)
Product Model	ACS5000	Perfect Harmony	TMdrive-MVG
Topology	5-L NPC	3-L	3-L
Power	1.7-24MVA	0.3-30MVA	0.18-10MVA
Voltage	2.3-13.8kV	4.16-6.9kV	3-11kV
Semiconductor	IGCT	LV/MV IGBT	GCT

(1) Toshiba, Mitsubishi and General Electric association

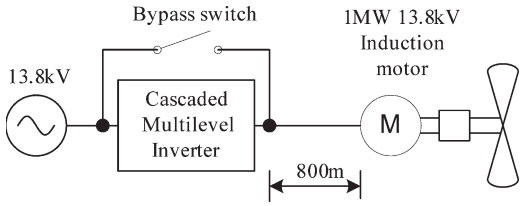


Fig. 13. Cascaded multilevel inverter application: Pumps and fans.

VI. APPLICATIONS

In this section, successful applications of CHB inverters are presented. Each one of these applications has particular requirements where cascaded inverters are well suited.

There are several suppliers of CHB inverters in the market of medium-voltage drives. Three of those suppliers and the technical characteristics of the CHB inverters they offer are shown in Table I.

A. Pumps and Fans

Pumps and fans are intensively used in almost all industry sectors. High-voltage high-power pumps and fans are used in water plants, oil and gas plants, cooling systems, geothermal and nuclear power plants, underground mining, furnaces and boilers, and so on.

The use of cascaded inverters to drive these devices could lead to an important efficiency improvement, because they typically run with variable speed at partial load. The use of variable speed drives, instead of dampers and throttling valves, to control the flow speed can reduce drastically the amount of power required. Fig. 13 shows an industrial fan application, where a 1-MW 13.8-kV induction motor is driven by a converter connected directly to the distribution system. The distance from the drive and the motor is about 800 m. The configuration shown in [62] presents the problem of voltage resonances at the motor terminals due to high-voltage variations over the long cables, requiring an LC filter between the converter and the load. However, if a CHB inverter is used, the voltage variations are greatly reduced, and the filter is also smaller if any.

B. STATCOM

One of the best suited applications for cascaded multilevel inverters is the power quality devices, like STATCOMs and universal power quality conditioners. These devices are connected directly to medium-voltage networks, as shown in Fig. 14, and do not require the injection of active power in a nominal operating point.

To accomplish with the first requirement, it is possible to connect as many inverters as required to reach the operating

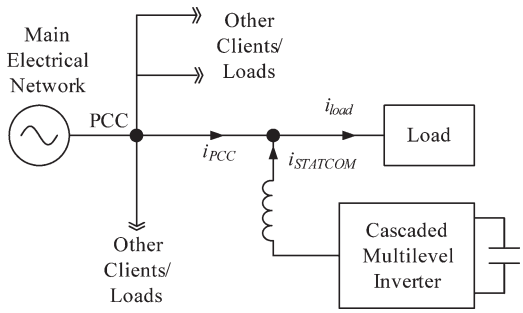


Fig. 14. Cascaded multilevel inverter application: STATCOM.

voltage, without the use of a transformer. The second requirement determines a simplification of the cascaded topology, which does not require a rectifier and input transformer stage, significantly reducing its costs. A combination of modulation and control techniques can provide floating and balanced dc voltages [63], [64].

Another alternative shown in [9] is to use photovoltaic cells to provide the floating dc voltages. The mentioned topology requires additionally a maximum power point tracker strategy to optimize the use of photovoltaic cells. However, in this case, it is possible to temporarily inject active power to the load.

C. Traction

Traction systems require a rectification stage of a high-voltage low-frequency ac power from the catenary and a fully controllable inversion stage to feed the traction motors. MMCs have been proposed to be used as an interface between the catenary voltage and low-voltage motor drives. The configuration used in [65] connects a single-phase line of 15 kV/16.7 Hz to three-phase 600-V induction motor drives using an MMC and a medium-frequency transformer. Classical cascaded multilevel inverters have also been proposed as a part of a power-quality compensator to reduce harmonics, reactive power, negative sequence, and the volatility of the load [66].

Applications of cascaded inverters on electric vehicles have been found in [67], where a back-to-back multilevel cascaded topology is proposed, and in [68], where a cascaded inverter with floating dc link is used as an inductorless boost inverter.

D. LNG Plant

The LNG plant presents a cyclic behavior during the year, motoring the turbine from the energy station in summer and reversing the power direction in winter when the energy consumption is higher. The use of a compressor directly connected to a gas turbine leads to an efficiency of 25%, due to the low efficiency of the turbine (approximately 30%). By using the scheme shown in Fig. 15, where the gas turbine has been replaced by a synchronous motor and a cascaded multilevel regenerative converter, the efficiency has been improved to 36%.

Due to the high power involved in this system (45 MW) and the bidirectional power flow, it is necessary to use a high-power converter with regeneration capability [29]. The cascaded

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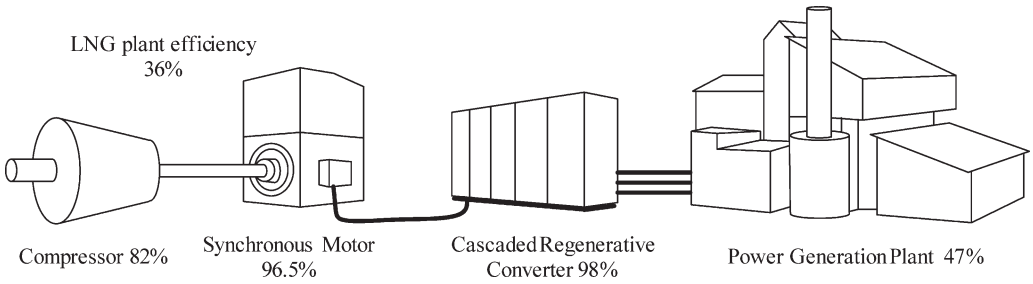


Fig. 15. Cascaded multilevel inverter application: LNG plant.

multilevel inverter emerges as the appropriate choice, considering also its extremely high availability. Cascaded inverters can minimize the maintenance effect in the production cycle, increasing the mean time between failures and, at the same time, reducing the maintenance work duration.

VII. Future Trends

Although this inverter topology has been established in the market of medium-voltage drives, there are some aspects that require further development and research. The first issue that needs attention is the efficiency improvement [69]. Several advances have been reported to reduce the switching losses using an optimal modulation technique; however, the conduction losses are far more critical due to the series connection of several semiconductors and high output currents. In order to reduce these losses, new advances in the semiconductor technology are expected. The use of floating cells could simplify the design of the input transformer or even eliminate it, which should reduce losses, cooling requirements, cost and volume.

To reach higher voltage levels is a challenge for semiconductor technology, increasing the blocking voltage and other related technologies like gate drivers and sensors.

Finally, the further increase on availability requires research on fault management, intelligent modularization, and the possibility of change modules and reconfigurations on the fly.

VIII. Comments and Conclusion

The cascaded multilevel inverters have evolved from a theoretical concept to real applications due to several remarkable features like a high degree of modularity, the possibility of connecting directly to medium voltage, high power quality, both input and output, high availability, and the control of power flow in the regenerative version.

This paper has reviewed the recent developments and applications of these inverters, including new proposed topologies, modulation techniques, and control strategies.

References

[1] J. Rodriguez, J.-S. Lai, and F. Z. Peng, "Multilevel inverters: A survey of topologies, controls, and applications," *IEEE Trans. Ind. Electron.*, vol. 49, no. 4, pp. 724–738, Aug. 2002.

[2] L. G. Franquelo, J. Rodriguez, J. I. Leon, S. Kouro, R. Portillo, and M. A. M. Prats, "The age of multilevel converters arrives," *IEEE Ind. Electron. Mag.*, vol. 2, no. 2, pp. 28–39, Jun. 2008.

[3] A. Nabae, I. Takahashi, and H. Akagi, "A new neutral-point-clamped PWM inverter," *IEEE Trans. Ind. Appl.*, vol. IA-17, no. 5, pp. 518–523, Sep./Oct. 1981.

[4] P. W. Hammond, "A new approach to enhance power quality for medium voltage AC drives," *IEEE Trans. Ind. Appl.*, vol. 33, no. 1, pp. 202–208, Jan./Feb. 1997.

[5] T. A. Meynard, H. Foch, P. Thomas, J. Courault, R. Jakob, and M. Nahrstaedt, "Multicell converters: Basic concepts and industry applications," *IEEE Trans. Ind. Electron.*, vol. 49, no. 5, pp. 955–964, Oct. 2002.

[6] J. Wen and K. M. Smedley, "Synthesis of multilevel converters based on single- and/or three-phase converter building blocks," *IEEE Trans. Power Electron.*, vol. 23, no. 3, pp. 1247–1256, May 2008.

[7] P. Lezana and G. Ortiz, "Extended operation of cascade multicell converters under fault condition," *IEEE Trans. Ind. Electron.*, vol. 56, no. 7, pp. 2697–2703, Jul. 2009.

[8] C. R. Baier, J. I. Guzman, J. R. Espinoza, M. A. Perez, and J. R. Rodriguez, "Performance evaluation of a multicell topology implemented with single-phase nonregenerative cells under unbalanced supply voltages," *IEEE Trans. Ind. Electron.*, vol. 54, no. 6, pp. 2969–2978, Dec. 2007.

[9] P. Flores, J. Dixon, M. Ortuzar, R. Carmi, P. Barriuso, and L. Moran, "Static Var compensator and active power filter with power injection capability, using 27-level inverters and photovoltaic cells," *IEEE Trans. Ind. Electron.*, vol. 56, no. 1, pp. 130–138, Jan. 2009.

[10] S. Mariethoz and A. Rufer, "Multisource DC-DC converter for the supply of hybrid multilevel converter," in *Conf. Rec. 41st IEEE IAS Annu. Meeting*, Oct. 2006, vol. 2, pp. 982–987.

[11] M. F. Escalante and J. J. Arellano, "Harmonics and reactive power compensation using a cascaded H-bridge multilevel inverter," in *Proc. IEEE Int. Symp. Ind. Electron.*, Jul. 2006, vol. 3, pp. 1966–1971.

[12] M. Carpaneto, M. Marchesoni, and L. Vaccaro, "A new cascaded multilevel converter based on NPC cells," in *Proc. IEEE ISIE*, Jun. 2007, pp. 1033–1038.

[13] Z. Wu, Y. Zou, and K. Ding, "Analysis of output voltage spectra in a hybrid diode-clamp cascade 13-level inverter," in *Proc. IEEE 36th PESC*, Jun. 2005, pp. 873–879.

[14] D. Kai, Z. Yunping, L. Lei, W. Zhichao, J. Hongyuan, and Z. Xudong, "Novel hybrid cascade asymmetric inverter based on 5-level asymmetric inverter," in *Proc. IEEE 36th Power Electron. Spec. Conf.*, Jun. 2005, pp. 2302–2306.

[15] X. Kou, K. A. Corzine, and M. W. Wielebski, "Overdistention operation of cascaded multilevel inverters," *IEEE Trans. Ind. Appl.*, vol. 42, no. 3, pp. 817–824, May/Jun. 2006.

[16] A. A. Sneineh and M.-Y. Wang, "Spectral analysis of hybrid capacitor-clamp cascade 13-level inverter," in *Proc. IEEE ICIT*, Dec. 2006, pp. 271–276.

[17] J. Dixon, A. A. Breton, F. E. Rios, J. Rodriguez, J. Pontt, and M. A. Perez, "High-power machine drive, using nonredundant 27-level inverters and active front end rectifiers," *IEEE Trans. Power Electron.*, vol. 22, no. 6, pp. 2527–2533, Nov. 2007.

[18] C. Rech and J. R. Pinheiro, "Hybrid multilevel converters: Unified analysis and design considerations," *IEEE Trans. Ind. Electron.*, vol. 54, no. 2, pp. 1092–1104, Apr. 2007.

[19] M. Perez, J. Rodriguez, J. Pontt, and S. Kouro, "Power distribution in hybrid multi-cell converter with nearest level modulation," in *Proc. IEEE ISIE*, Jun. 2007, pp. 736–741.

[20] R. R. Astudillo, D. Ruiz-Caballero, M. S. Ortmann, and S. A. Mussa, "New symmetrical hybrid multilevel DC-AC converters," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2008, pp. 1916–1922.

[21] M. Rotella, G. Penailillo, J. Pereda, and J. Dixon, "PWM method to eliminate power sources in a nonredundant 27-level inverter for machine

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drive applications," *IEEE Trans. Ind. Electron.*, vol. 56, no. 1, pp. 194–201, Jan. 2009.
[22] Z. Du, B. Ozpineci, and L. M. Tolbert, "Modulation extension control of hybrid cascaded H-bridge multilevel converters with 7-level fundamental frequency switching scheme," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2007, pp. 2361–2366.
[23] S. Vazquez, J. I. Leon, L. G. Franquelo, J. J. Padilla, and J. M. Carrasco, "DC-voltage-ratio control strategy for multilevel cascaded converters fed with a single DC source," *IEEE Trans. Ind. Electron.*, vol. 56, no. 7, pp. 2513–2521, Jul. 2009.
[24] H. Liu, L. M. Tolbert, S. Khomfoi, B. Ozpineci, and Z. Du, "Hybrid cascaded multilevel inverter with PWM control method," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2008, pp. 162–166.
[25] S. Lu and K. A. Corzine, "Advanced control and analysis of cascaded multilevel converters based on P-Q compensation," *IEEE Trans. Power Electron.*, vol. 22, no. 4, pp. 1242–1252, Jul. 2007.
[26] M. Veenstra and A. Rufer, "Control of a hybrid asymmetric multilevel inverter for competitive medium-voltage industrial drives," *IEEE Trans. Ind. Appl.*, vol. 41, no. 2, pp. 655–664, Mar./Apr. 2005.
[27] G.-J. Su, "Multilevel DC-link inverter," *IEEE Trans. Ind. Appl.*, vol. 41, no. 3, pp. 848–854, May/Jun. 2005.
[28] P. Lezana and J. Rodriguez, "Mixed multicell cascaded multilevel inverter," in *Proc. IEEE ISIE*, Jun. 2007, pp. 509–514.
[29] J. R. Rodriguez, J. Pontt, M. Perez, P. Lezana, and P. W. Hammond, "High power synchronous machine fed by a cascaded regenerative inverter," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2008, pp. 1–7.
[30] P. Lezana, C. A. Silva, J. Rodriguez, and M. A. Perez, "Zero-steady-state-error input-current controller for regenerative multilevel converters based on single-phase cells," *IEEE Trans. Ind. Electron.*, vol. 54, no. 2, pp. 733–740, Apr. 2007.
[31] P. Lezana, J. Rodriguez, D. Rojas, and J. Pontt, "Novel cell based on reduced single-phase active front end for multicell converters," in *Proc. 31st IEEE IECON*, Nov. 2005, pp. 733–738.
[32] P. Lezana, J. Rodriguez, and D. A. Oyarzun, "Cascaded multilevel inverter with regeneration capability and reduced number of switches," *IEEE Trans. Ind. Electron.*, vol. 55, no. 3, pp. 1059–1066, Mar. 2008.
[33] A. J. Watson, P. W. Wheeler, and J. C. Clare, "A complete harmonic elimination approach to DC link voltage balancing for a cascaded multilevel rectifier," *IEEE Trans. Ind. Electron.*, vol. 54, no. 6, pp. 2946–2953, Dec. 2007.
[34] S. Kouro, P. Lezana, M. Angulo, and J. Rodriguez, "Multicarrier PWM with DC-link ripple feedforward compensation for multilevel inverters," *IEEE Trans. Power Electron.*, vol. 23, no. 1, pp. 52–59, Jan. 2008.
[35] S. Inoue and H. Akagi, "A bidirectional isolated DC–DC converter as a core circuit of the next-generation medium-voltage power conversion system," *IEEE Trans. Power Electron.*, vol. 22, no. 2, pp. 535–542, Mar. 2007.
[36] M. Glinka and R. Marquardt, "A new AC/AC multilevel converter family," *IEEE Trans. Ind. Electron.*, vol. 52, no. 3, pp. 662–669, Jun. 2005.
[37] M. Hagiwara and H. Akagi, "PWM control and experiment of modular multilevel converters," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2008, pp. 154–161.
[38] P. E. Melin, J. R. Espinoza, N. R. Zargari, L. A. Moran, and J. I. Guzman, "A novel multi-level converter based on current source power cell," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2008, pp. 2084–2089.
[39] J. I. Leon, S. Vazquez, S. Kouro, L. G. Franquelo, J. M. Carrasco, and J. Rodriguez, "Unidimensional modulation technique for cascaded multilevel converters," *IEEE Trans. Ind. Electron.*, vol. 56, no. 8, pp. 2981–2986, Aug. 2009.
[40] J. Rodriguez, S. Kouro, J. Rebolledo, and J. Pontt, "A reduced switching frequency modulation algorithm for high power multilevel inverters," in *Proc. IEEE 36th Power Electron. Spec. Conf.*, Jun. 2005, pp. 867–872.
[41] Z. Du, L. M. Tolbert, and J. N. Chiasson, "Active harmonic elimination for multilevel converters," *IEEE Trans. Power Electron.*, vol. 21, no. 2, pp. 459–469, Mar. 2006.
[42] Z. Du, L. M. Tolbert, J. N. Chiasson, and B. Ozpineci, "Reduced switching-frequency active harmonic elimination for multilevel converters," *IEEE Trans. Ind. Electron.*, vol. 55, no. 4, pp. 1761–1770, Apr. 2008.
[43] M. Perez, S. Kouro, J. Rodriguez, and B. Wu, "Modified staircase modulation with low input current distortion for multicell converters," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2008, pp. 1989–1994.
[44] M. G. Hosseini Aghdam, S. H. Fathi, and G. B. Gharehpetian, "Elimination of harmonics in a multi-level inverter with unequal DC sources using the homotopy algorithm," in *Proc. IEEE ISIE*, Jun. 2007, pp. 578–583.
[45] V. G. Agelidis, A. I. Balouktsis, and M. S. A. Dahidah, "A five-level symmetrically defined selective harmonic elimination PWM strategy: Analysis and experimental validation," *IEEE Trans. Power Electron.*, vol. 23, no. 1, pp. 19–26, Jan. 2008.
[46] M. Angulo, P. Lezana, S. Kouro, J. Rodriguez, and B. Wu, "Level-shifted PWM for cascaded multilevel inverters with even power distribution," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2007, pp. 2373–2378.
[47] R. Gupta, A. Ghosh, and A. Joshi, "Switching characterization of cascaded multilevel-inverter-controlled systems," *IEEE Trans. Ind. Electron.*, vol. 55, no. 3, pp. 1047–1058, Mar. 2008.
[48] Y.-M. Park, H.-S. Yoo, H.-W. Lee, M.-G. Jung, S.-H. Lee, C.-D. Lee, S.-B. Lee, and J.-Y. Yoo, "A simple and reliable PWM synchronization and phase-shift method for cascaded H-bridge multilevel inverters based on a standard serial communication protocol," in *Conf. Rec. 41st IEEE IAS Annu. Meeting*, Oct. 2006, vol. 2, pp. 988–994.
[49] O. Lopez, J. Alvarez, J. Doval-Gandoy, and F. D. Freijedo, "Multilevel multiphase space vector PWM algorithm," *IEEE Trans. Ind. Electron.*, vol. 55, no. 5, pp. 1933–1942, May 2008.
[50] A. Das, K. Sivakumar, R. Ramchand, C. Patel, and K. Gopakumar, "A combination of hexagonal and 12-sided polygonal voltage space vector PWM control for IM drives using cascaded two-level inverters," *IEEE Trans. Ind. Electron.*, vol. 56, no. 5, pp. 1657–1664, May 2009.
[51] A. K. Gupta and A. M. Khambadkone, "A space vector modulation scheme to reduce common mode voltage for cascaded multilevel inverters," *IEEE Trans. Power Electron.*, vol. 22, no. 5, pp. 1672–1681, Sep. 2007.
[52] A. K. Gupta and A. M. Khambadkone, "A general space vector PWM algorithm for multilevel inverters, including operation in overmodulation range," *IEEE Trans. Power Electron.*, vol. 22, no. 2, pp. 517–526, Mar. 2007.
[53] Z. Cheng and B. Wu, "A novel switching sequence design for five-level NPC/H-bridge inverters with improved output voltage spectrum and minimized device switching frequency," *IEEE Trans. Power Electron.*, vol. 22, no. 6, pp. 2138–2145, Nov. 2007.
[54] V. Oleschuk, F. Profumo, A. Tenconi, R. Bojoi, and A. M. Stankovic, "Cascaded three-level inverters with synchronized space-vector modulation," in *Conf. Rec. 41st IEEE IAS Annu. Meeting*, Oct. 2006, vol. 2, pp. 595–602.
[55] J. I. Leon, S. Vazquez, A. J. Watson, L. G. Franquelo, P. W. Wheeler, and J. M. Carrasco, "Feed-forward space vector modulation for single-phase multilevel cascaded converters with any DC voltage ratio," *IEEE Trans. Ind. Electron.*, vol. 56, no. 2, pp. 315–325, Feb. 2009.
[56] S. Alepuz, S. Busquets-Monge, J. Bordonau, J. A. Martinez-Velasco, C. A. Silva, J. Pontt, and J. Rodriguez, "Control strategies based on symmetrical components for grid-connected converters under voltage dips," *IEEE Trans. Ind. Electron.*, vol. 56, no. 6, pp. 2162–2173, Jun. 2009.
[57] H. Miranda, V. Cardenas, G. Espinosa-Perez, and D. Noriega-Pineda, "Multilevel cascade inverter with voltage and current output regulated using a passivity based controller," in *Conf. Rec. 41st IEEE IAS Annu. Meeting*, Oct. 2006, vol. 2, pp. 974–981.
[58] M. Perez, R. Ortega, and J. R. Espinoza, "Passivity-based PI control of switched power converters," *IEEE Trans. Control Syst. Technol.*, vol. 12, no. 6, pp. 881–890, Nov. 2004.
[59] M. A. Perez, P. Cortes, and J. Rodriguez, "Predictive control algorithm technique for multilevel asymmetric cascaded H-bridge inverters," *IEEE Trans. Ind. Electron.*, vol. 55, no. 12, pp. 4354–4361, Dec. 2008.
[60] P. Zanchetta, D. B. Gerry, V. G. Monopoli, J. C. Clare, and P. W. Wheeler, "Predictive current control for multilevel active rectifiers with reduced switching frequency," *IEEE Trans. Ind. Electron.*, vol. 55, no. 1, pp. 163–172, Jan. 2008.
[61] C.-C. Hua, C.-W. Wu, and C.-W. Chuang, "A digital predictive current control with improved sampled inductor current for cascaded inverters," *IEEE Trans. Ind. Electron.*, vol. 56, no. 5, pp. 1718–1726, May 2009.
[62] J. Rodriguez, J. Pontt, C. Silva, R. Musalem, P. Newman, R. Vargas, and S. Fuentes, "Resonances and overvoltages in a medium-voltage fan motor drive with long cables in an underground mine," *IEEE Trans. Ind. Appl.*, vol. 42, no. 3, pp. 856–863, May/Jun. 2006.
[63] J. A. Barrena, L. Marroyo, M. A. R. Vidal, and J. R. T. Apraiz, "Individual voltage balancing strategy for PWM cascaded H-bridge converter-based STATCOM," *IEEE Trans. Ind. Electron.*, vol. 55, no. 1, pp. 21–29, Jan. 2008.
[64] Y. Liu, A. Q. Huang, W. Song, S. Bhattacharya, and G. Tan, "Small-signal model-based control strategy for balancing individual DC capacitor voltages in cascade multilevel inverter-based STATCOM," *IEEE Trans. Ind. Electron.*, vol. 56, no. 6, pp. 2259–2269, Jun. 2009.
[65] M. Glinka, "Prototype of multiphase modular-multilevel-converter with 2 MW power rating and 17-level-output-voltage," in *Proc. IEEE 35th PESC*, 2004, vol. 4, pp. 2572–2576.

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[66] L. Zhou, Q. Fu, X. Li, and C. Liu, "A novel multilevel power quality compensator for electrified railway," in *Proc. IEEE 6th IPEMC*, May 2009, pp. 1141–1147.
[67] A. Dell'Aquila, M. Liserre, V. G. Monopoli, and C. Cecati, "Design of a back-to-back multilevel induction motor drive for traction systems," in *Proc. IEEE 34th Power Electron. Spec. Conf.*, Jun. 2003, vol. 4, pp. 1764–1769.
[68] Z. Du, B. Ozpineci, L. M. Tolbert, and J. N. Chiasson, "DC–AC cascaded h-bridge multilevel boost inverter with no inductors for electric/hybrid electric vehicle applications," *IEEE Trans. Ind. Appl.*, vol. 45, no. 3, pp. 963–970, May 2009.
[69] T. Kinjo, K. Takao, Y. Tanaka, K. Sung, and H. Ohashi, "Quantitative study on operation frequency limitation of multi-level high voltage power converter equipped with Si-IEGT and SiC-PiN diode," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2008, pp. 2909–2913.



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